Smart Health Monitoring Bracelet Prottype - Real-timet Rate & SpO_2 Measurement with a Kosler Neon Interface

- Continuous HR Tracking
- SPO_2 Measurement

- Activity & Sleep Analysis
- Long-lasting Battery Life

Introduction to the Protorype

- Welcome to the future of continuous, non-invasive health monitoring. This protorype is designed as an accesible, wearable smart bracelet.

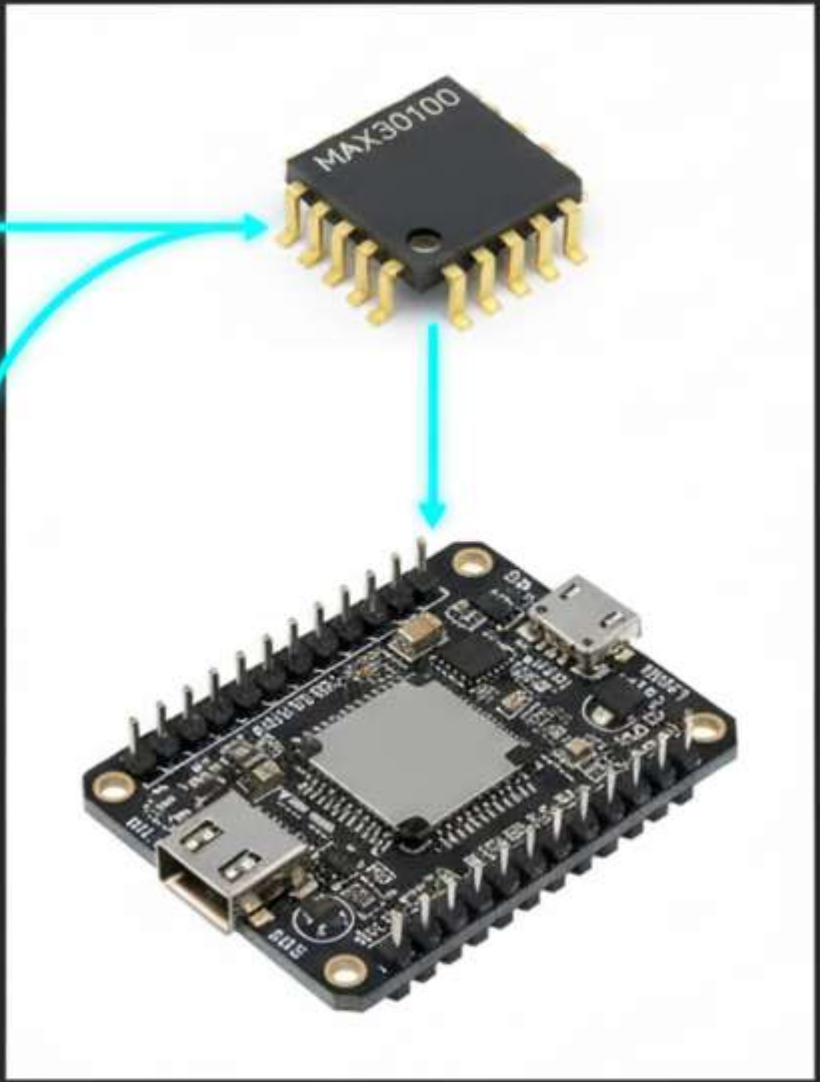
- Its primary function is the highly accurate measurement two critical physiologgical parameters: heart rate (BPM) and blood oxgen saturation, SpO_2.

- The system completely eliminates the need of cumbrorrose wiring, seamsmlting realime-retime biometric readings wirelesy to a conmected computer or dashboard.

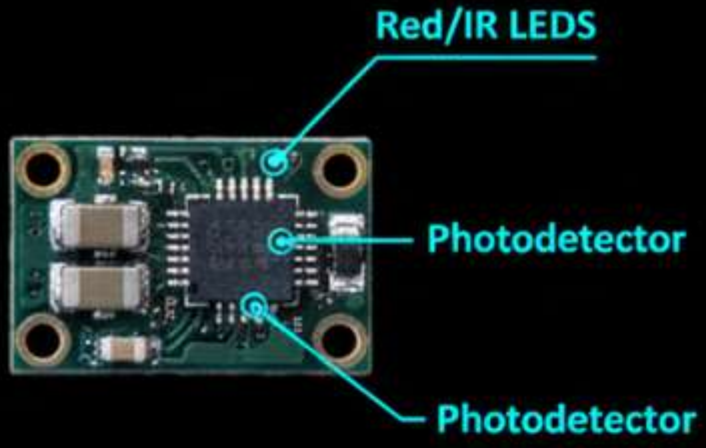
- It serves as foundational step toward predictive persona! healthcare, putting critical data riata right at the user fingetrips.

Core Hardware Components

- The architecture of the protorype reliss on an optimized, dual-component hardware stack designed for reliability and low power consumption.
- First, the MAX30100 sensor acts the primary biometic data collection module, capturing raw physliogical signals signals with clinical-grade methology.
- Second, the NodeMU ESP8266 board serves the central cant aray pratairg, take raw analotg data, processing, and handling all wireless communications.
- This compact and highly efficient design ensurs that the physical footrrint remains small enough for integration into wearable IOT form factor.



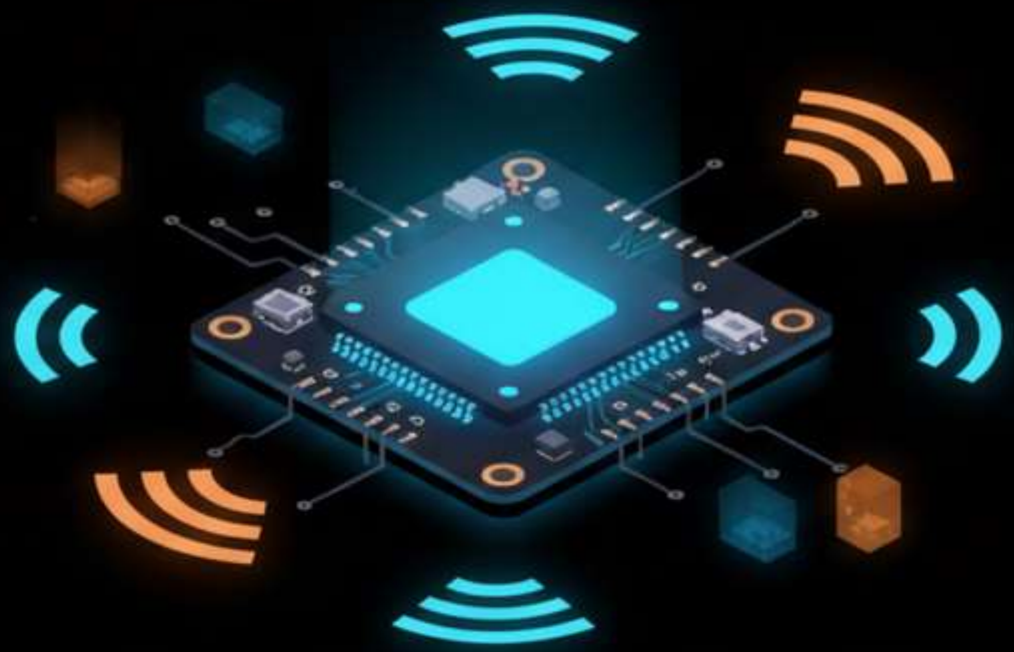
Deep Dive: The MAX30100 Sensor



- The MAX30100 is specialized, purpose-built sensor designed specfically for integrated for integrated pulse oximtetry and heart rate monitoring.
- It operates by combining high-eficiency Red and Infarraed (IR LEDs with ultra-sensitive photedector to read light absorption through skin.
- An advanced, buitin analogg signal processing unit filters out ambient light and motionc | motion artifacts, ensuring the captured pulse signals are clean and highly accurate.
- Its ultra-low power operation makes it the ideal coice for battery-powered weaable health devices.

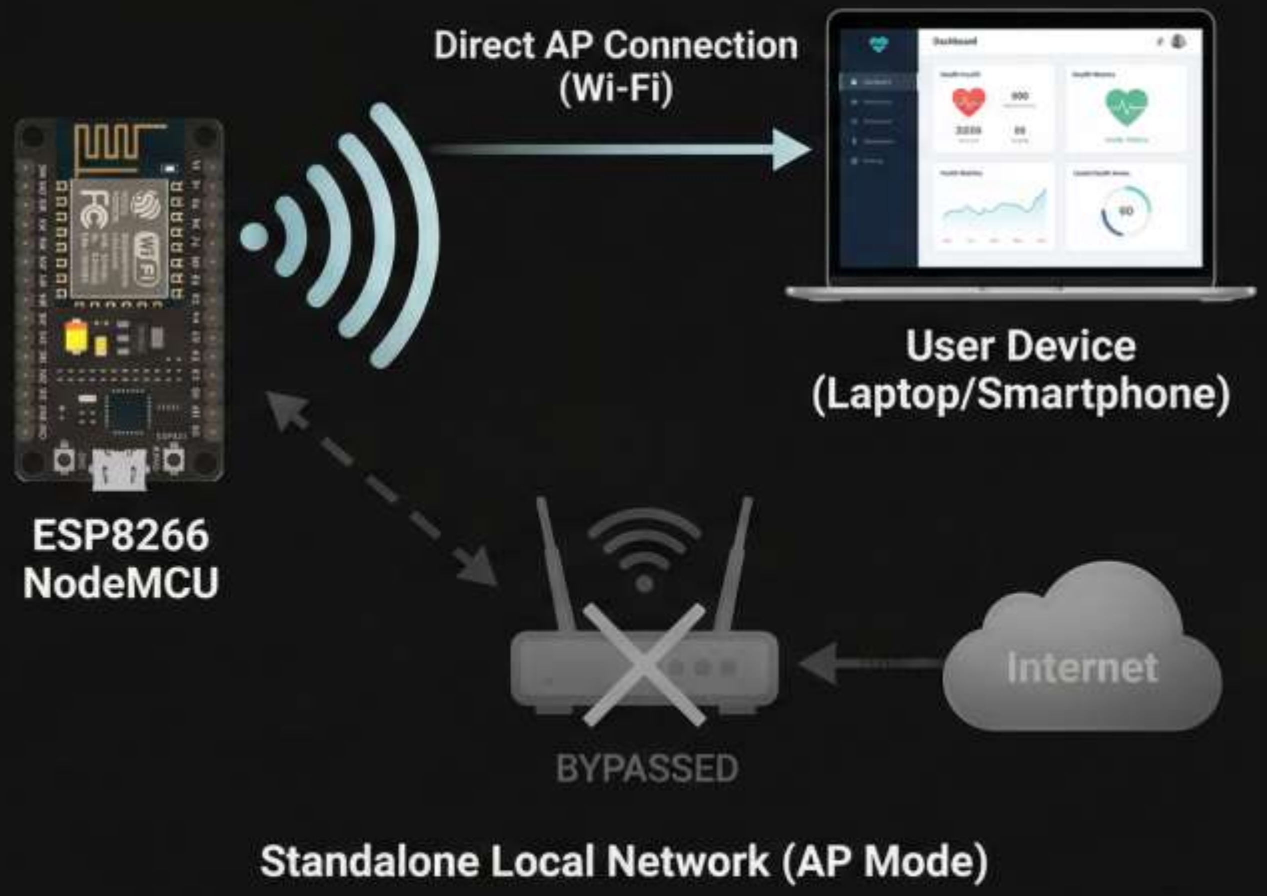
Deep Dive: NodeMCU ESP8266

- Acts als powerful mincrocuolor brain, brain, managing computation & connecivity.
- Rapidaly reads, filters, and processes analog & digital data data from MAX30100 sensor.
- Fully integrated Wi-Fi module for seamless wireless communication.
- Hosts & broadcasts user interface without secordary computing device



Wireless Network Strategy (AP Mode)

- NodeMCU configures its own standalone local network known as Access Point (AP) mode.
- Smart bracelet operates completely independently and does not require a local internet router to function.
- A user simply turns on the device, and a computer or smartphone connects directly to the bracelet's broadcasted Wi-Fi network.
- This decentralized approach guarantees data privacy, zero internet latency, and the ability to monitor health metrics in remote locations without Wi-Fi.

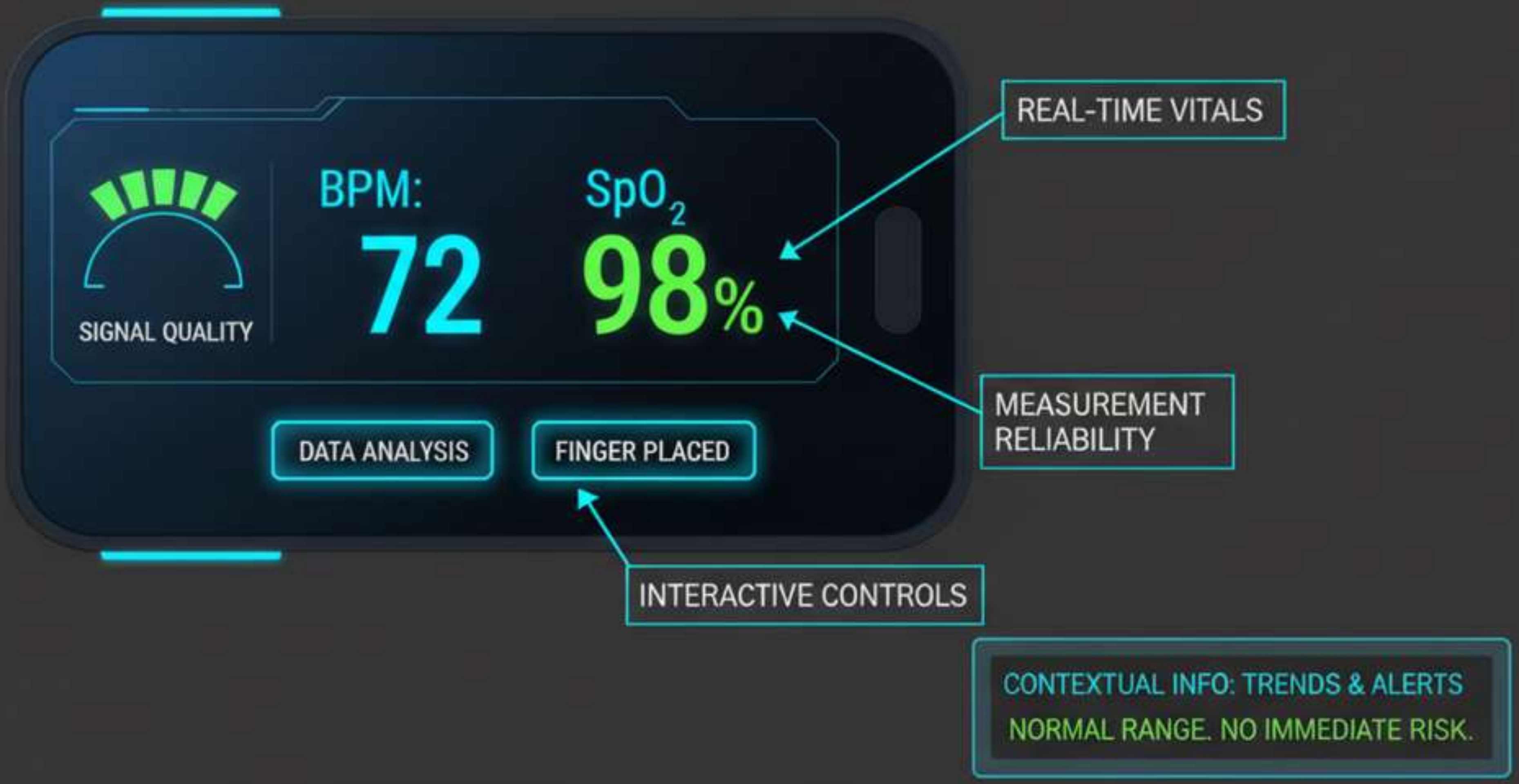


Web Interface Design

- Lightweight local web interface served directly from protorpe's ESP866 memory.
- Striking "Kosler Neon" aesthetic with vibrant accents reduces eye strain.
- Organized & responsive layout for immedate data interpation.
- Bridges sophisticate medical data and intutive consumer technology

Interface Elements and Outputs

Monsserrt Bold;
Robotot Regular



Operational Workflow: Setup and Connection

- **Step 1: Sensor Contact**
  Finger on MAX30100 Pad

Step 2: Data Processing
NodeMCU Reads Pulse & O_2

Step 3: UI Access - PC
Connects to Local IP

- The NodeMCU instant boots finger on the illuminated pad a oibiaases sensor, and begins continuiouls reading the rading the raw pulse and oxygen values.

- Simatastonecly, the device establishes its local W-Fi Access Point (AP). user connects their PC to this network and navigtes and designated local IP address to open a UI.

Operational Workflow: The Analysis Phase

- Step 4: Click "Data Analysis" to verify sensor placement.
  Step 5: Confirm with "Finger Placed" for diagnostic mode
- Step 6: Timed recording (seconds to minutes)
  for biometric data

Displaying the Results

Physological Condition
Normal

Estimated Health Risk
Low Risk

Current Metrics Analysis
Stable

- Upon the conclusion of to recording session, the interface smoothly snnoothly transtions to neon rectangulal notification box featuring a promment checkmark.
- The system instantt generates an intepretative text result detailing user's current physiiogical condition based on the captured metrics.
- An estimated health risk level level, displyed, utilizing color-coded text (e. g), green for healthy, yellow, for caution) to provide imedate, easily, digestible feedback.

System Diagnostic Categories

Normal

Both Heart Rate and SPO_2 readings fall comfortably within standard, healthy medical ranges.

Needs Monitoring

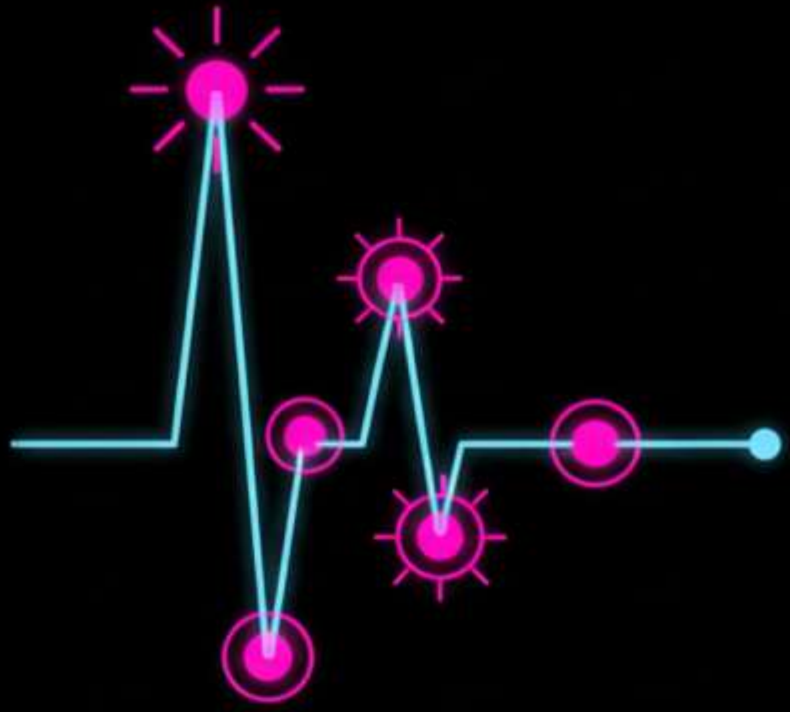
The system detects slight deviations or minor inossitenie in the pulse or pusor oxygen levels, suggesting the user should rest and retest shortly.

Requires Medical Attention

Significant abhormal readings are detected (such danigrossly low SPO_2 or extreme taxchradia/bradycars), triggering and: the urgent health alert.

The intelligent backend categorize the gathrred data into three primary health states tolés thelp users understand their current condition.

DATA ANALYSIS & MEDICAL CONTEXT

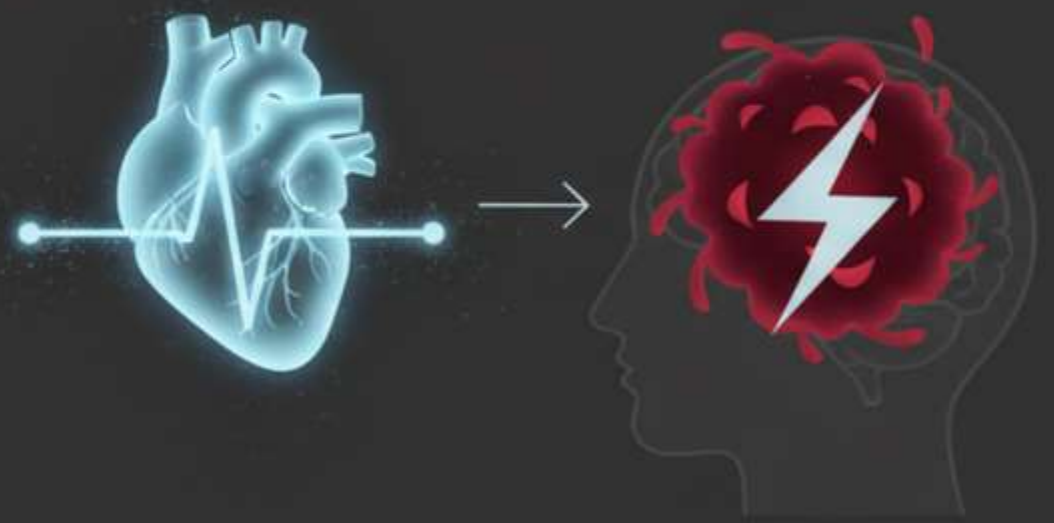


- The primary focus of our data analysis algorithm is identify subtle, early warning indicators of potential health crisies.
- It continouuls scans the pulse stream to detect severe irreulalites, missed herised beats, or abnormal heart rate variabiity the variabiity that the user might not physicaly feel.
- Simalitatienly, it rigurouls monitors for signficant blood oxygen saturation (SpO_2), which can ean r sign be early sign of respiratory distress or cardosucutal inffiiciency.
- By providing context to raw numbers, the system acts as an proactive health health companion rather just a passive tracking device.

System Limitations & Disclaimer

- While highly advanced, this prototype is NOT a definitive diagnostic tool for conditions like strokes or advanced heart diseases.
- Provided data must NOT be considered a viable replacement for certified professional medical devices or the opinion of a licensed physician.
- This iteration DOES NOT measure blood pressure or provide a full 12-lead ECG reading.
- Its purpose is STRICTLY INFORMATIONAL—acting as a screening tool to encourage timely professional medical consultations.

The Link Between Pulse and Stroke Risk



- Why monitor the pulse so closely? Irregular pulse patterns are a primary indicator of undiangosed arrihiamiss, such Atrial Fibrlation (AFib.
- AFib causes the upper chamers of heart to beat irregually, which significantly increases of probabiity of blood pooling, clotting, and ultimatly traveing to the brain to cause stroke.

  pavnabols stroke.
- Continuous, daily monitoring of pulse reguality as a vital, non-invasive measu
- Detecting these hidden arrthramiss early is a onla of most effective ways of mitigate the the risk of stroke.

The Value of Early Warning Systems

- The true value of this prototype lies in its ability to identify abormal biometric markers at a a very early, pre-symptomontic stage.
- It directly empowers users to take charge of their health, providing the data necessary to seek out a professional evaluation evaluation before a minor condition severely worltenss.
- Smart monitoring devices like this bridge a crucial gap between btween casual daily lifestyle monitoring and clinical, professional healthcare.
- They democrratie access a essential health metrics, transforming reactive healthcare into proactive health management.

Future Enhancements

Integration with Mobile Ecosstems

Transtioning from a local web AP to dedicated IOS and Android applications for broader accessiblity.

Data Tracking & History

Implementing cloud or localiized database storage to allow usrs track their historical data and share reports with their physicians

Hardware Upgrades

Enhancing sensor shielding and employing machine calibration dation to improve raw accuracy in diverse lighting conditions

Algoriathicc Refiement

Upgrading the software to utilize advanced AI anomliy detection, signficanly improving te recogntion of specific arrhihamis like AFib.

Summary and Conclusion

- Developed a highly capable, cost-effective, and entirely wireless health monitoring IoT solution.
- Merged robust hardware sensing (MAX30100 & NodeMCU) with an intuitive, visually engaging 'Kosler Neon' web interface for an accessible user experience.
- Created a powerful, proactive tool designed to enhance personal health awareness and daily safety.
- Represents a significant step forward in bringing critical biometric screening straight to the consumer.

Thank You

Robor you for your time, attention, and interest in the future of wearable health technology.
The floor now now open for any questions, feedback, or a live system demonstration.